Inversion

Version 4.0

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Contents

Introduction

Inversion is an advanced tool that can be used to solve some geometry problems. It is a geometric transformation that totally changes the figure, such as converting circles to straight lines and vice versa. Sometimes the problem may be simplified by performing an inversion to the figure. Although this is not a necessary tool to solve some IMO problems, it is always better if we have more possible approaches to solve the same problem. We are going to introduce this special tool in this short note.

In section 1, we first discuss some basic properties of inversion. Then in the remaining sections 2 to 4, we go through different usages of inversion and go through some common tricks involved when performing an inversion.

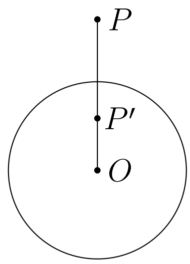
1 Properties of Inversion

In this section, we first introduce a transformation called inversion, which is the main tool that we will use in the whole note. Official solutions to MO problems usually do not require inversion. However, it is a common alternative solution to many geometry problems, especially the hard ones. Therefore, those who are familiar with all the basic stuff in geometry should learn inversion and understand its usages. One may regard this as a more advanced tool that can be useful occasionally.

Definition 1.1. Let O be a point on the plane and let k be a nonzero constant. The *inversion*(反演) at centre O with ratio k is the transformation that sends every point $P \neq O$ to the unique point P' on the line OP such that $OP \cdot OP' = k$ (in directed lengths).

Sometimes we also define the image of O to be a point at infinity. Unlike the projective plane, inversion is usually defined on the Möbius plane, where there is only one single point at infinity through which every line passes. Conversely, the image of this point at infinity is defined to be O. Under this definition, an inversion maps the Möbius plane bijectively to itself. Also, if P' is the inversion image of P, then P is the inversion image of P'.

When $k > 0$, P and P' lie on the same side of O on the line OP. Choose $r > 0$ such that $k = r^2$. We usually define the transformation to be the inversion with respect to(w.r.t.) the circle with centre O and radius r. In that case, P lies inside the circle if and only if(iff) P' lies outside the circle, and P is mapped to itself iff P lies on the circle. In view of this, inversion is commonly known as the reflection about a circle.

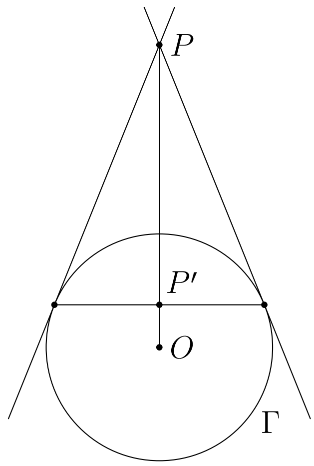


When $k < 0$, O lies between P and P'. This can be regarded as an inversion with ratio $|k|$ together with the reflection in O. In most cases, we should only consider inversions with positive ratios, though sometimes this is also useful.

Throughout the entire note, we use X' to denote the image of a point X under an inversion unless otherwise specified. Note that one should mention this clearly when presenting a solution using inversion.

We now go through some important properties of this new transformation one by one.

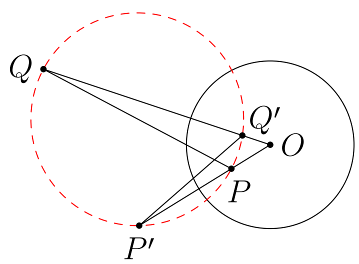
Proposition 1.1. Let Γ be a circle. The inversion image of P w.r.t. Γ lies on the polar of P w.r.t. Γ. In particular, if P lies outside Γ, then P' is the midpoint of the two contact points of the tangents from P.



Proof. This follows from proposition 3.1(c) of **Projective Geometry 4.0**. □

This gives a simple way to find the image of a point outside Γ. Also, this gives a connection between inversion and projective geometry.

Proposition 1.2. Consider an inversion with centre O. If P and Q are two points such that O, P, Q are not collinear, then P, P', Q, Q' are concyclic. Also, we have $\triangle OPQ \sim \triangle OQ'P'$.



Proof. By definition, $OP \cdot OP' = OQ \cdot OQ'$. This implies P, P', Q, Q' are concyclic. The result $\triangle OPQ \sim \triangle OQ'P'$ follows from this group of concyclic points. □

As a corollary of the pair of similar triangles, we have $\angle OPQ = \angle OQ'P'$. This gives us a simple way to relate the angles before and after the transformation. We shall find this result useful in working on some problems later on.

When using inversion, the first major task we need to do is to find out the images of everything in the original figure. This is not limited to the points which are labelled in the question statement. For example, if there is a segment AB in the figure, we also need to know the images of all points on this segment in order to plot the image of AB. Clearly, the result needs not be the segment $A'B'$ since we did not mention that inversion preserves the shape of some objects (which is not true in general). More generally, we would like to know the image of a curve (such as a line or a circle) under an inversion. This means the collection of the images of all points on that curve. We have the following simple result.

Proposition 1.3. Consider an inversion with centre O. Then the following hold.

(a) The image of a line ℓ passing through O is ℓ itself.

(b) The image of a line ℓ not passing through O is a circle passing through O. In particular, the tangent at O to the image circle is parallel to ℓ.

(c) The image of a circle Ω passing through O is a line not passing through O. In particular, the image line is parallel to the tangent at O to Ω.

(d) The image of a circle Ω not passing through O is a circle not passing through O. In particular, the centres of the two circles and O are collinear.

Proof.

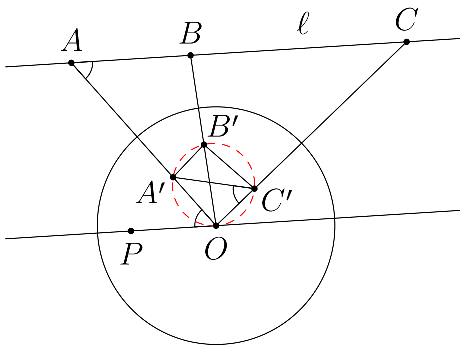
(a) This is obvious because for every point P on ℓ, the image P' lies on $OP = \ell$, and every point on ℓ is the image of some point on ℓ since inversion is bijective. Note that the result does not mean every point on ℓ remains unchanged. In fact, some pairs of points are swapped with each other.

(b) Suppose A, B, C lie on a line ℓ not passing through O. By proposition 1.2, both A, A', B, B' and B, B', C, C' are concyclic. Therefore, B' is the Miquel point of $\triangle BA'C'$ w.r.t. $\triangle OCA$, and hence O, A', B', C' are concyclic. As C varies, its image C' always lies on $(OA'B')$. So the images of the points on ℓ lie on a single circle passing through O. When C tends to the point at infinity in either direction, C' tends to O from either side. Due to continuity, the image of ℓ is the whole circle.

   Let P be any point on the line passing through O and parallel to ℓ. As A, A', C, C' are concyclic, we have

$$\measuredangle A'OP = \measuredangle OAC = \measuredangle A'C'O.$$

   So this is the tangent at O to the image circle.

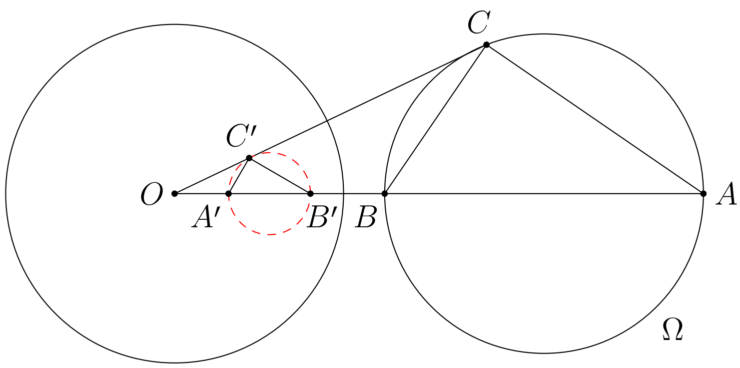


(c) This follows from (b) and the fact that $(P')' = P$ for every point P.

(d) Let AB be a diameter of Ω such that O lies on the line AB. Consider any point C on Ω different from A and B. We claim that C' lies on the circle with diameter $A'B'$. Indeed, using proposition 1.2, we find that

$$\measuredangle A'C'B' = \measuredangle OC'B' - \measuredangle OC'A' = \measuredangle CBO - \measuredangle CAO = \measuredangle BCA = \frac{\pi}{2}.$$

This proves the claim, and so C' always lies on a fixed circle. Using continuity, again it is easy to see that the image of Ω is the whole circle with diameter $A'B'$. In addition, the midpoint of AB (the centre of Ω), the midpoint of $A'B'$ (the centre of the image) and O are collinear by construction. However, it does not mean that the midpoint of AB is mapped to the midpoint of $A'B'$.



□

In some cases, we can find the images of some curves explicitly. For example, if a line ℓ cuts the circle of inversion Γ at two distinct points A and B, then the image of ℓ is (OAB) according to (b) because A and B are unchanged under the transformation. Similarly, if ℓ is tangent to Γ at A, then its image is the circle internally tangent to Γ at A and passing through O.

Similarly, if a circle Ω intersects Γ at two distinct points A and B, then its image circle, Ω and Γ are coaxial, having AB as their common radical axis.

Lastly, we mention a result that allows us to find the distances between two image points under an inversion.

Proposition 1.4. Consider an inversion with centre O and radius r. Then

$$A'B' = \frac{r^2}{OA \cdot OB} \cdot BA.$$

Proof. If O, A, B are collinear, then (in directed lengths)

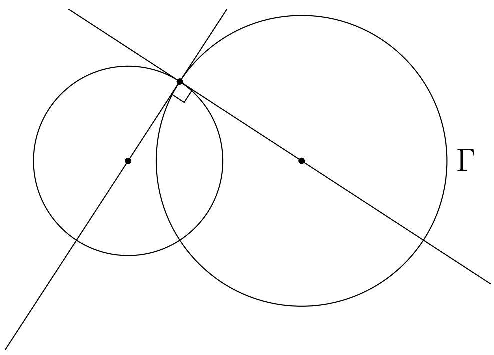
$$\frac{r^2}{OA \cdot OB} \cdot BA = \frac{r^2}{OA \cdot OB} \cdot (BO + OA) = -\frac{r^2}{OA} + \frac{r^2}{OB} = -OA' + OB' = A'B'.$$

If O, A, B are not collinear, then $\triangle OAB \sim \triangle OB'A'$ by proposition 1.2. This implies $\dfrac{A'B'}{BA} = \dfrac{OB'}{OA}$, and hence

$$A'B' = \frac{OB'}{OA} \cdot BA = \frac{r^2}{OA \cdot OB} \cdot BA.$$

$\square$

In addition, it is known that inversion preserves the angle of intersection of two curves (which means the angle between the tangents to the two curves). This general fact is not that useful for our purpose. The most common usage of this fact is that inversion preserves the orthogonality of two curves. In particular, a circle is invariant under an inversion w.r.t. Γ iff it is orthogonal to Γ. The meaning of orthogonal circles is indicated in the figure below.

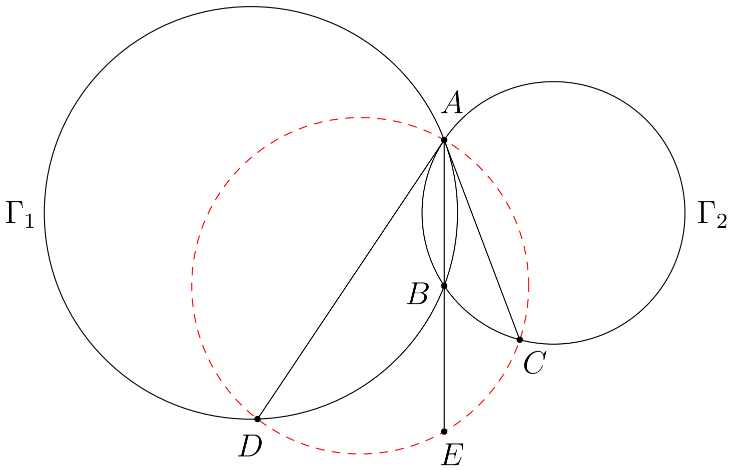


2 Changing the Figures

How do we use inversion to solve a geometry problem? A simple idea is that by applying an inversion to the original figure, different points and curves are mapped to new points and new curves. This gives us a new figure to work on, which could be totally different from the original one. In other words, we can transform the original problem to a new one by using inversion. This could simplify the whole problem in some cases if the inversion is carefully chosen.

To apply an inversion, we first need to determine the centre and the ratio. In some cases, the ratio is less important, and so the major task is to determine the centre of inversion. In view of proposition 1.3, one way to simplify the problem is to choose the intersection point of some circles to be the centre, since after the inversion, all these circles will become straight lines. Certainly straight lines are simpler than circles.

Problem 2.1. Two circles Γ_1 and Γ_2 intersect at distinct points A and B. The tangent at A to Γ_1 meets Γ_2 at C. The tangent at A to Γ_2 meets Γ_1 at D. Let E be the point such that B is the midpoint of AE. Prove that A, C, E, D are concyclic.



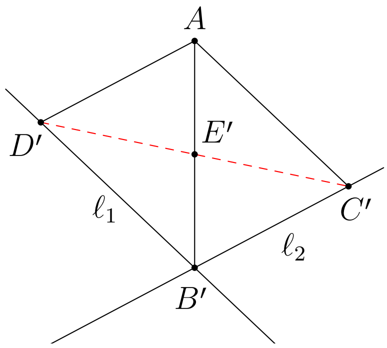
Analysis. There are three circles in this problem, Γ_1, Γ_2 and the circle passing through A, C, E, D (assuming the result is correct). All these circles pass through the same point A. According to our discussion, it is the best to choose A to be the centre of inversion if we are going to use this transformation. We use this problem to demonstrate what we need to do when applying an inversion.

The first task is to find the images of all points and objects. In view of our construction, we may begin with Γ_1 and Γ_2. By proposition 1.3(c), they are mapped to two straight lines ℓ_1 and ℓ_2 not passing through A. What can we say about ℓ_1 and ℓ_2? We mainly need to know whether the lines intersect or not. Indeed, since both Γ_1 and Γ_2 pass through B, their images ℓ_1 and ℓ_2 must pass through B'. Therefore, we first draw two lines which intersect each other at a point B'. We do not even need to know the angle between the two lines as it is unimportant in this problem. (But actually we know that this angle is the angle between Γ_1 and Γ_2 by a property of inversion, which means $\angle CAD$.)

The next step is to plot the point C'. Of course it lies on ℓ_2. Apart from that, recall that AC is tangent to Γ_1 at A. This implies AC' (the image of AC) has exactly one intersection point with ℓ_1. This sounds useless. However, be careful that this intersection point is indeed the image of A, which is the point at infinity. In other words, AC' is parallel to ℓ_1. Similarly, AD' is parallel to ℓ_2.

Lastly, we can locate point E' on the line AB' by using the definition. In this way, we successfully change the original figure to a new one. Now, what are we going to prove? Again by proposition 1.3(c), A, C, E, D are concyclic iff C', E', D' are collinear. This is the thing we need to prove.

Solution.



Apply an inversion with centre A and an arbitrary ratio k. The images of Γ_1 and Γ_2 are two lines ℓ_1 and ℓ_2 respectively. The intersection point of ℓ_1 and ℓ_2 is B'. Since AC is tangent to Γ_1, AC' is parallel to ℓ_1. Similarly, AD' is parallel to ℓ_2. Also, since

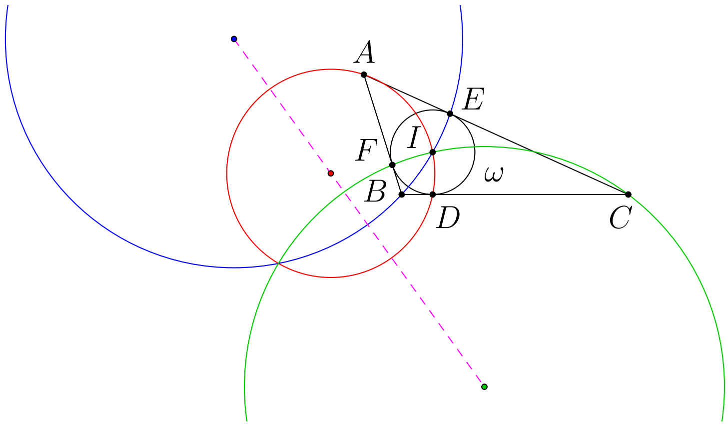
$$AE' = \frac{k}{AE} = \frac{k}{2AB} = \frac{AB'}{2},$$

E' is the midpoint of AB'.

The problem is now rephrased as follows. There is a parallelogram $AD'B'C'$, and E' is the midpoint of AB'. We are going to show that C', E', D' are collinear. But this is obvious because the diagonals of a parallelogram bisect each other! Once we have known that C', E', D' are collinear, we can use proposition 1.3 to deduce that A, C, E, D are concyclic. This completes the proof. □

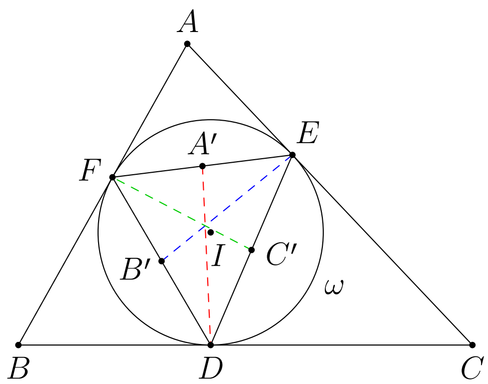
Remarks. Notice how different the original problem and the new problem are. This shows the usage of inversion. In some cases, the problem is turned into some well-known facts after an inversion. Even if the figure is not greatly simplified, it is still good to obtain an entirely new problem to work on, since we have more ways to tackle the problem.

Problem 2.2. The incircle ω of $\triangle ABC$ touches the sides BC, CA, AB at D, E, F respectively. Prove that the circumcentres of $\triangle AID$, $\triangle BIE$ and $\triangle CIF$ are collinear.



Analysis. The problem does not seem to be easy. However, noting that the three circles in the conclusion all pass through I, it is natural to consider an inversion at centre I. Since there is already a circle ω in the figure with centre I, we may consider the inversion w.r.t. ω. Indeed, performing an inversion w.r.t. the incircle of a triangle is a common step in solving problems in triangle geometry. One will find that the problem is again trivialized by applying this inversion.

Solution.



Consider the inversion w.r.t. ω. Note that D, E, F are invariants as they lie on ω, and so there is no need to draw another new figure. We simply need to know what are the images of A, B, C and the three circles in the problem.

Firstly, by proposition 1.1, A, B, C are mapped to the midpoints of EF, FD, DE respectively. For (AID), as it passes through the centre I of inversion, its image is a straight line passing through A' and D (the images of A and D) by proposition 1.3. In other words, the image of (AID) is the line $A'D$. Similarly, the images of (BIE) and (CIF) are $B'E$ and $C'F$ respectively.

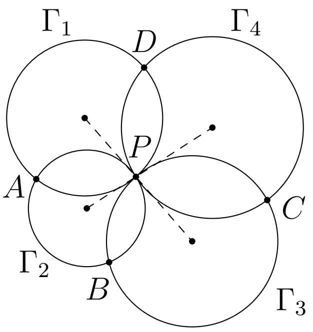
To show that the centres of $(AID), (BIE), (CIF)$ are collinear, it is the same as proving the circles are coaxial since they pass through a common point I. In other words, we have to show that they intersect at a point different from I. Equivalently, we have to show that their images intersect at a point different from the point at infinity. (Strictly speaking, there is another possibility that all three circles are tangent at I, which means their images are parallel lines.)

Indeed, note that $A'D, B'E, C'F$ are the medians of $\triangle DEF$, and hence they must be concurrent at the centroid of $\triangle DEF$. This simple fact solves the whole problem immediately. $\square$

Remarks. There are many facts in triangle geometry that can be proved using the same inversion. For example, one can use this to show that the line joining the circumcentre and the incentre of a triangle is the Euler line of the intouch triangle.

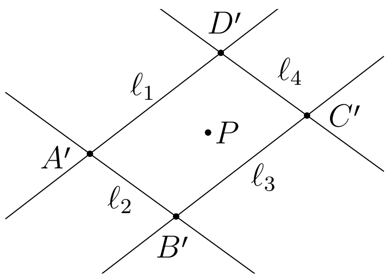
Problem 2.3. (IMO Shortlist 2003 G4) Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be distinct circles such that Γ_1, Γ_3 are externally tangent at P, and Γ_2, Γ_4 are externally tangent at the same point P. Suppose that Γ_1 and Γ_2; Γ_2 and Γ_3; Γ_3 and Γ_4; Γ_4 and Γ_1 meet at A, B, C, D, respectively, and that all these points are different from P. Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}.$$



Analysis. The same trick and idea can already be used to solve a shortlist problem like this. The original figure consists of four circles, which is quite complicated. However, by applying an inversion at centre P, all circles are mapped to straight lines. This greatly simplifies the figure. This time we try to use the distance formula to compare the lengths before and after inversion.

Solution.



Consider an arbitrary inversion with centre P and ratio k. Again, the images of $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ are some straight lines $\ell_1, \ell_2, \ell_3, \ell_4$ respectively. Since each ℓ_j is parallel to the tangent at P to Γ_j, and Γ_1, Γ_3 are externally tangent at P, we have $\ell_1 // \ell_3$. Similarly, $\ell_2 // \ell_4$.

As A lies on both Γ_1 and Γ_2, its image A' is the intersection of ℓ_1 and ℓ_2. Similarly, the images of A, B, C, D form a parallelogram $A'B'C'D'$ enclosed by $\ell_1, \ell_2, \ell_3, \ell_4$.

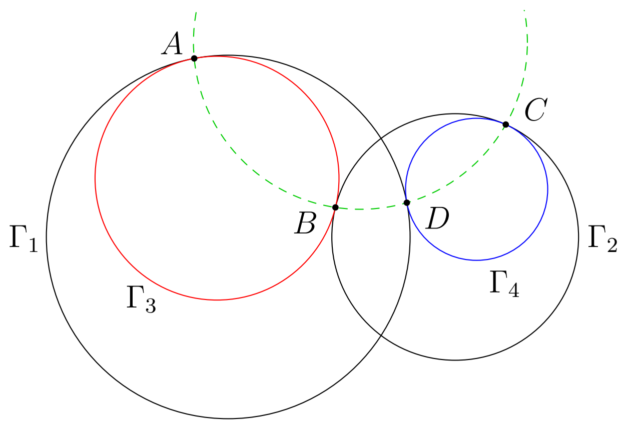
It remains to convert the side lengths in the conclusion to the lengths in the new figure. For example, by proposition 1.4, we have $AB = \dfrac{PA \cdot PB}{k} \cdot A'B'$. Similarly,

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{\left(\dfrac{PA \cdot PB}{k} \cdot A'B'\right)\left(\dfrac{PB \cdot PC}{k} \cdot B'C'\right)}{\left(\dfrac{PA \cdot PD}{k} \cdot A'D'\right)\left(\dfrac{PD \cdot PC}{k} \cdot D'C'\right)} = \frac{PB^2}{PD^2} \cdot \frac{A'B' \cdot B'C'}{A'D' \cdot D'C'}.$$

As $A'B'C'D'$ is a parallelogram, we have $A'B' = D'C'$ and $B'C' = A'D'$. Therefore, the second fraction is equal to 1. The result follows readily. $\square$

Remarks. In view of these problems, when there are many circles (and lines) passing through the same point in the figure, we may consider an inversion with centre being that intersection point.

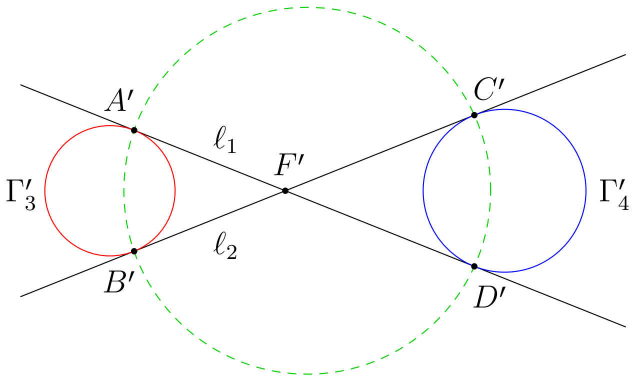
Problem 2.4. Two circles Γ_1 and Γ_2 intersect at two distinct points. A circle Γ_3 lies inside Γ_1, and is internally tangent to Γ_1 at A and externally tangent to Γ_2 at B. A circle Γ_4 lies inside Γ_2, and is internally tangent to Γ_2 at C and externally tangent to Γ_1 at D. Prove that A, B, D, C are concyclic or collinear.



Analysis. Beginners might find this kind of problems hard because there are too many circles. Now we have introduced a tool that helps us transform circles to straight lines, so it is a good idea to use the same strategy here.

Unlike the previous problems, the four circles do not pass through the same point. We can at most invert at the intersection point of two circles to transform both of them to straight lines. This is not bad as we can still reduce the number of circles. To retain the symmetry, we may choose an intersection point of Γ_1 and Γ_2 as the centre of inversion.

Solution.



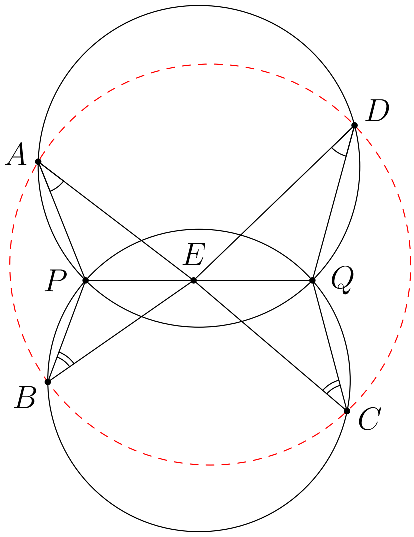
Let E and F be the intersection points of Γ_1 and Γ_2. Consider an inversion at centre E. Clearly, Γ_1 and Γ_2 are mapped to two straight lines ℓ_1 and ℓ_2 which intersect at F'. Since Γ_3 does not pass through E, its image Γ_3' is a circle by proposition 1.3(d). As Γ_3 is tangent to Γ_1 and Γ_2 at A and B respectively, Γ_3' is tangent to ℓ_1 and ℓ_2 at A' and B' respectively. Similarly, Γ_4' is tangent to ℓ_1 and ℓ_2 at D' and C' respectively. In view of the configuration, $\angle A'F'B'$ and $\angle C'F'D'$ should be vertical opposite angles. (For example, when Γ_4 varies continuously, it never intersects Γ_3, and hence cannot pass through A and B. This shows Γ_4' must lie in the region which is opposite to the region that Γ_3' lies in.)

The rest is simple. Note that the centres of Γ_3' and Γ_4' lie on the internal angle bisector of $\angle A'F'B'$. Therefore, A' and B' are symmetric about this angle bisector, C' and D' are symmetric about this angle bisector. It follows that $A'B'D'C'$ is an isosceles trapezoid by symmetry, and hence A', B', D', C' are concyclic. Again by proposition 1.3, A, B, D, C are concyclic or collinear (depending on whether E lies on $(A'B'D'C')$ or not). $\square$

Remarks. Using the same inversion, one can show that AB and CD always pass through the same fixed point when Γ_3 and Γ_4 vary (whose image is the reflection of E about the internal angle bisector of $\angle A'F'B'$).

We have seen several applications of inversion to convert circles to straight lines. Next, we shall see how to use inversion to simplify some strange angle conditions.

Problem 2.5. (IMO Shortlist 2008 G3) Let $ABCD$ be a convex quadrilateral and let P and Q be points in $ABCD$ such that $PQDA$ and $QPBC$ are cyclic quadrilaterals. Suppose that there exists a point E on the line segment PQ such that $\angle PAE = \angle QDE$ and $\angle PBE = \angle QCE$. Show that the quadrilateral $ABCD$ is cyclic.

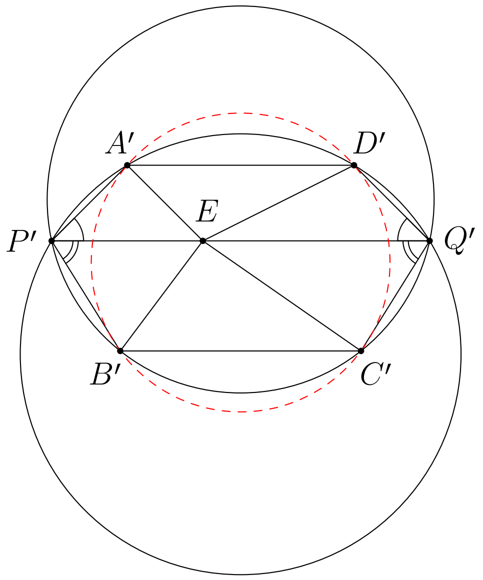


Analysis. There are three circles in the figure (including $(ABCD)$). Will we apply an inversion to turn some of the circles to straight lines? This does not seem to simplify the problem because the main barrier in solving this problem is the strange angle conditions. This time we shall see how to use inversion to interpret this kind of conditions.

To do so, let us first recall how angles are changed under an inversion. Suppose we apply an inversion at centre O. For any distinct points A, B, C different from O, we have $\angle A'OB' = \angle AOB$ and $\angle OA'B' = \angle OBA$, while $\angle A'B'C'$ in general does not have some simple relations with the interior angles of $\triangle ABC$ (there are still some ways to express $\angle A'B'C'$ in terms of some other angles before inversion, which we will discuss in later problems). In view of this, it is easier to handle some angles where the centre of inversion is one of the labels in the angles. As we are going to handle $\angle EAP$, $\angle EDQ$, $\angle EBP$ and $\angle ECQ$ in this problem, it is obvious that the best choice of the centre is E. We shall see how this simplifies everything.

Solution. Consider an arbitrary inversion at centre E. Note that P', E, Q' are collinear. By proposition 1.2, we have $\angle A'P'E = \angle PAE$ and $\angle D'Q'E = \angle QDE$. Using the given condition, we deduce $\angle A'P'E = \angle D'Q'E$. This is a much nicer condition than the given one because it simply means two adjacent interior angles of the quadrilateral $D'A'P'Q'$ are equal. In addition, we also know that $D'A'P'Q'$ is cyclic since $DAPQ$ is cyclic. It follows from $\angle A'P'Q' = \angle D'Q'P'$ that $D'A'P'Q'$ is an isosceles trapezoid.

Similarly, we also know that $B'C'Q'P'$ is an isosceles trapezoid. Both isosceles trapezoids $D'A'P'Q'$ and $B'C'Q'P'$ share the same axis of symmetry (the perpendicular bisector of $P'Q'$). Therefore, $A'B'C'D'$ is another isosceles trapezoid by symmetry. In particular, it is cyclic, and hence $ABCD$ is cyclic. $\square$

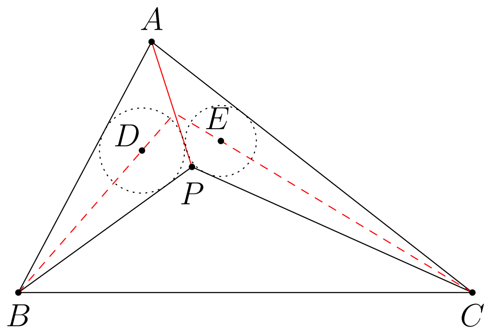


Remarks. Amazingly, this seemingly hard problem is also turned to a trivial problem after applying a suitable inversion. In the solution, we demonstrate how inversion can be used to flip an angle to another position for comparison.

Problem 2.6. (IMO Shortlist 1996 G2/Problem 2) Let P be a point inside triangle ABC such that

$$\angle APB - \angle ACB = \angle APC - \angle ABC.$$

Let D, E be the incentres of triangles APB, APC, respectively. Show that AP, BD, CE meet at a point.

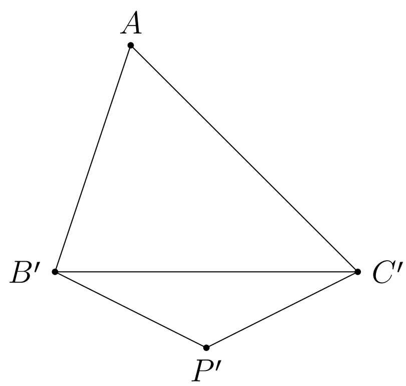


Analysis. The idea is the same as the previous problem. We wish to apply an inversion to move the four angles in the given condition to different positions, so that it is easier to add or subtract these angles. Since all angles share the common label A, we can consider an inversion at centre A.

However, how do we find the images of D and E? This is a tedious task if we wish to do so. Finding the image of the incircle of $\triangle ABP$ is fine, but the centre of the image circle needs not be D'. We can only rely on the definition or some basic properties to work out a construction for D' (for example, AD' is still the angle bisector of $\angle B'AP'$, and D' lies on (DPP'), etc.). To make things simple, we first try to find an equivalent way to interpret the problem so that it is not necessary for us to find the exact position of D'.

For example, the final conclusion basically means the internal angle bisectors of $\angle ABP$ and $\angle ACP$ intersect on AP. This formulation does not require the points D and E at all. But still it is not easy to find the images of these angle bisectors as they become circles. Indeed, we can further simplify the problem. Using the angle bisector theorem, it suffices to prove that $\frac{BA}{BP} = \frac{CA}{CP}$, since this means the two angle bisectors cut AP in the same ratio. In this final formulation, we can work entirely on the four points A, B, C, P, and we do not bother to find the images of D and E.

Solution.



Consider an inversion at centre A. Then we have $\angle APB = \angle AB'P'$, etc., and so

$$\angle APB - \angle ACB = \angle AB'P' - \angle AB'C' = \angle P'B'C',$$
$$\angle APC - \angle ABC = \angle AC'P' - \angle AC'B' = \angle P'C'B'.$$

(Note that $\angle AB'P' > \angle AB'C'$ as $\angle APB = \angle PAC + \angle ACB + \angle CBP > \angle ACB$.) Therefore, the given angle condition actually means $\angle P'B'C' = \angle P'C'B'$. Thus, $P'B' = P'C'$.

Suppose AP meets BD and CE at X and Y respectively. By the angle bisector theorem, we have

$$\frac{XA}{XP} = \frac{BA}{BP} \quad \text{and} \quad \frac{YA}{YP} = \frac{CA}{CP}.$$

As $X = Y$ iff $\frac{XA}{XP} = \frac{YA}{YP}$, it suffices to prove $\frac{BA}{BP} = \frac{CA}{CP}$ in order to show that AP, BD, CE are concurrent. By proposition 1.2, we have $\triangle ABP \sim \triangle AP'B'$. Therefore,

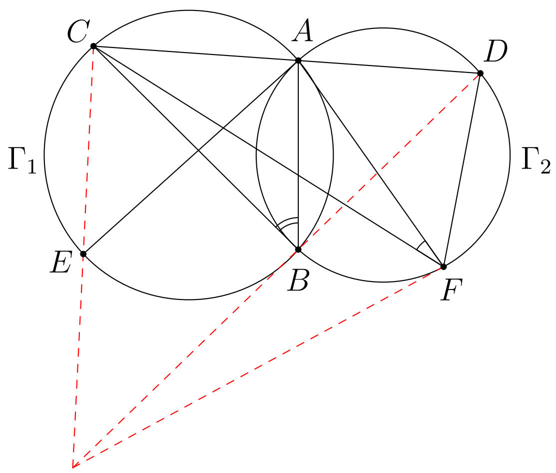
$$\frac{BA}{BP} = \frac{P'A}{P'B'}.$$

Similarly, $\dfrac{CA}{CP} = \dfrac{P'A}{P'C'}$. But then

$$\frac{BA}{BP} = \frac{P'A}{P'B'} = \frac{P'A}{P'C'} = \frac{CA}{CP}.$$

This completes the proof. □

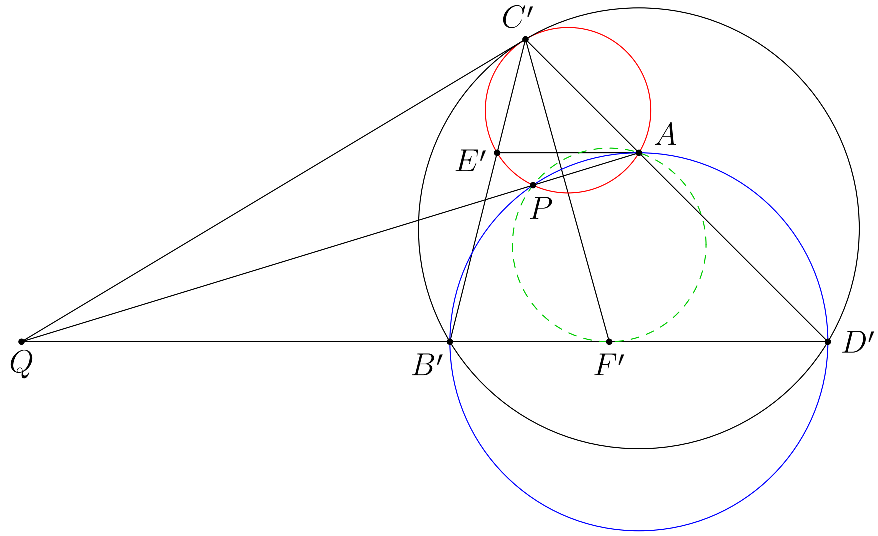
Problem 2.7. (Iranian Geometry Olympiad 2021 Advanced Problem 2) Two circles Γ_1 and Γ_2 meet at two distinct points A and B. A line passing through A meets Γ_1 and Γ_2 again at C and D respectively, such that A lies between C and D. The tangent at A to Γ_2 meets Γ_1 again at E. Let F be a point on Γ_2 such that F and A lie on different sides of BD, and $2\measuredangle AFC = \measuredangle ABC$. Prove that the tangent at F to Γ_2, and lines BD and CE are concurrent.



Analysis. Again, the relation $2\measuredangle AFC = \measuredangle ABC$ is the most annoying condition. However, after flipping the angles under an inversion at centre A, the two angles are put together, and this essentially mean an angle bisector condition. If one is familiar with this idea, it is natural to apply such an inversion. We are also able to transform both circles to straight lines.

Solution. Consider an inversion at centre A. The images of Γ_1 and Γ_2 are straight lines not passing through A. Therefore, C', E', B' are collinear, and B', F', D' are collinear. Also, C', A, D' are still collinear. Thus, we now have a $\triangle C'B'D'$, and points E', F', A on the sides $C'B', B'D', D'C'$ respectively.

There are some more conditions that we know on these six points. Firstly, since AE is tangent to Γ_2, this implies AE' has exactly one intersection point with the line $D'B'$. But this intersection point actually means the point at infinity. Thus, $AE' // D'B'$. Secondly, the condition $2\measuredangle AFC = \measuredangle ABC$ means $2\measuredangle AC'F' = \measuredangle AC'B'$ in the new figure. Equivalently, $C'F'$ is the internal angle bisector of $\measuredangle B'C'D'$.



Next, let us work out the thing to be proved. The lines CE and BD are mapped to $(AC'E')$ and $(AB'D')$ respectively. The tangent at F to Γ_2 is mapped to a circle passing through A and F' which is tangent to $B'D'$ (the image of Γ_2) at F'. We are going to show that these three circles intersect at a point different from A. Indeed, let P be the second intersection of $(AC'E')$ and $(AB'D')$. We shall show that (APF') is tangent to $B'D'$ at F'.

It does not seem easy to chase the angles related to $\triangle APF'$. Therefore, using power maybe a better option. Let AP meet $B'D'$ at Q. We are going to compare $QP \times QA$ and QF'^2. To use the condition $E'A // B'D'$, one way is to consider a homothety between $\triangle C'E'A$ and $\triangle C'B'D'$. In particular, $(AC'E')$ (which is a circle we need to consider) is homothetic to $(D'C'B')$ at centre C', and hence they are tangent at C'. Now we have three circles $(AC'E')$, $(AB'D')$ and $(D'C'B')$. By considering their radical axes, we see that AP, $B'D'$ and the common tangent at C' to $(AC'E')$ and $(D'C'B')$ are concurrent at Q.

It is well-known that $QF' = QC'$. For example, this follows from

$$\angle QF'C' = \angle F'C'D' + \angle C'D'F' = \angle B'C'F' + \angle QC'B' = \angle QC'F'.$$

Therefore,

$$QF'^2 = QC'^2 = QP \times QA,$$

showing that QF' is tangent to (APF'). This completes the proof. □

Remarks. This time the problem is not trivialized after inversion. We still need a lot of geometric knowledge to figure out this nice solution after the transformation. But then the good thing is that we have an entirely new problem to work on, which looks like a more familiar problem in triangle geometry to us. In particular, the double angle condition becomes a really simple condition after the inversion. This is one situation which we may think of using inversion.

In fact, there is a possibility that $P = A$, which means the three circles are tangent at A, or the tangent at F to Γ_2, BD and CE are parallel.

There are two main usages of inversion that we have seen in this section.

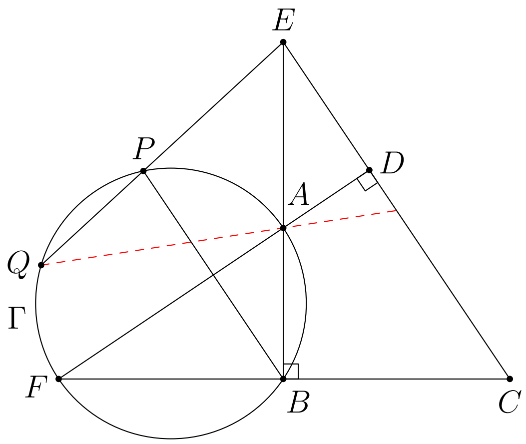
- changing circles to straight lines (problems 2.1, 2.2, 2.3, 2.4)
- moving angles (problems 2.5, 2.6, 2.7)

3 Swapping the Points

In the previous section, we mainly use inversion to change the entire figure to work on a new problem. In this section, we would like to see how the new figure obtained after an inversion can be compared with the original figure to tell us more information about the original setting. To do so, we wish to keep as many points in the figure as possible when we apply the transformation. This does not mean that they must be invariants. It is good enough if some pairs of points simply swap with each other.

To do so, we have to look for a centre O for which there are many pairs of points, say A and A', such that $OA \cdot OA'$ is a constant. In order that $OA \cdot OA' = OB \cdot OB'$, the points A, A', B, B' should be concyclic (or collinear). Therefore, one strategy is to look for a cyclic quadrilateral, and use the intersection point of their opposite sides or diagonals as the centre of inversion.

Problem 3.1. Let $ABCD$ be a quadrilateral with $\angle B = \angle D = 90°$. The lines AB and CD meet at E, the lines AD and BC meet at F. The line passing through B and parallel to CD meets the circumcircle Γ of $\triangle ABF$ again at P. The line EP meets Γ again at Q. Prove that the line QA bisects the segment CE.



Analysis. In view of our discussion, it is a good idea to perform an inversion at centre E. By choosing a suitable ratio, we can swap P and Q, and swap A and B since A, B, Q, P are concyclic. In addition, A, B, C, D are also concyclic. Therefore, the same inversion swaps C and D as well. Let us see what we can deduce by using this inversion.

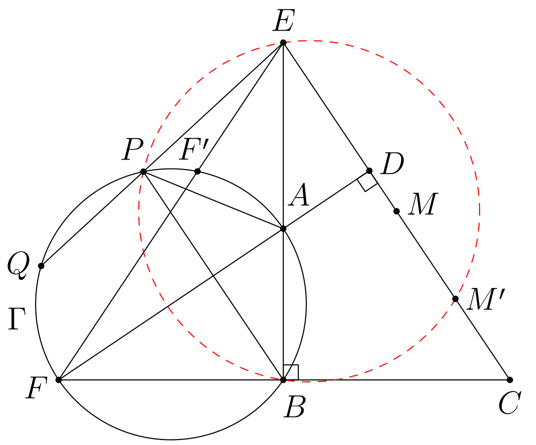
Solution. Consider the inversion with centre E and ratio $EP \cdot EQ$. Since

$$EP \cdot EQ = EA \cdot EB = ED \cdot EC,$$

the following pairs of points are swapped: $(P, Q), (A, B), (D, C)$. Also, F' is the second intersection point of EF and Γ for the same reason (which is not needed anyway). Let M be the midpoint of CE. By definition,

$$EM' = \frac{EP \cdot EQ}{EM} = \frac{ED \cdot EC}{EM} = 2ED.$$

In other words, D is the midpoint of EM'.



To show that Q, A, M are collinear, it is the same as showing P, B, M', E are concyclic since P, B, M' are the images of Q, A, M respectively. Recall that $PB // EC$. Therefore, we have to show $EPBM'$ is an isosceles trapezoid. It is clear that AD is the perpendicular bisector of EM'. It remains to show that it is also the perpendicular bisector of PB. As $PB // EC$, we have $PB \perp AD$. Also,

$$\angle APB = \angle AFB = 90^\circ - \angle FAB = \angle ABP.$$

This implies $AP = AB$, and hence AD is the perpendicular bisector of PB. We are done by the above discussion. $\square$

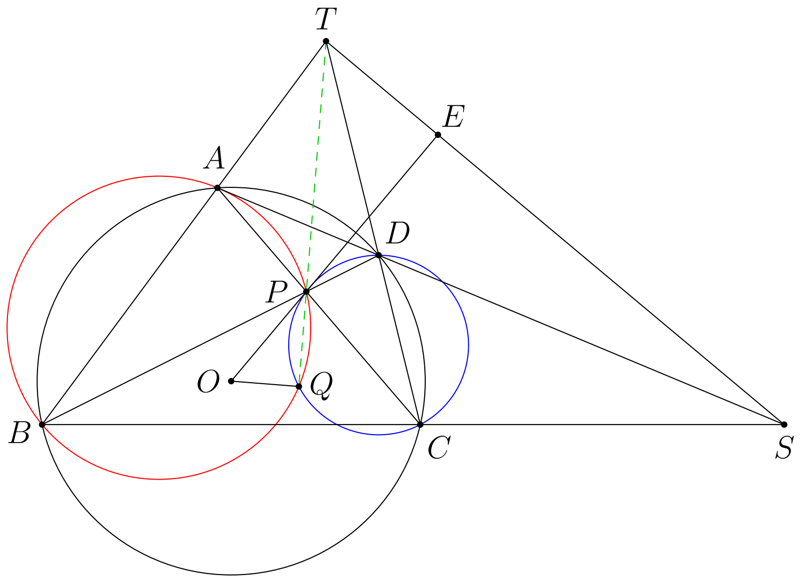
Remarks. Following this strategy, we only need to add a few points to the figure. Nevertheless, the whole problem is changed to another form, from proving collinear points to proving concyclic points. This allows us to attempt the same problem in a completely different way, even though the figure does not change by too much.

Problem 3.2. (CMO 1992 Problem 4) A convex quadrilateral $ABCD$ is inscribed in circle O. Its diagonals AC and BD intersect at P. The circumcircles of $\triangle ABP$, $\triangle CDP$ intersect at P and another point Q, where O, P, Q are pairwise distinct. Prove that $\angle OQP = 90^\circ$.

Analysis. Using an inversion at centre P seems to be a good idea. We can swap A and C, as well as B and D. At the same time, the circles (ABP) and (CDP) are turned to some straight lines. Following this idea, we would naturally define the intersection point T of AB and CD. But then this is insufficient to prove the result at once.

The right angle in the final result may remind us Brokard's theorem, which is related to cyclic quadrilaterals and orthocentres (hence perpendicular lines). Thus, we also define the intersection point S of AD and BC. Brokard's theorem says that each of O, P, S, T is the orthocentre of the triangle formed by the other three points. In addition, we further know that ST is the polar of P w.r.t. $\odot O$, etc. This is closely related to proposition 1.1, and so it should also be useful to consider an inversion w.r.t. $\odot O$.

Solution.



Let S be the intersection of AD and BC, let T be the intersection of AB and CD, and let E be the intersection of ST and OP. It is well-known that ST is the polar of P w.r.t. $\odot O$. This implies $OP \perp ST$. In addition, by proposition 1.1, we see that E is the image of P under the inversion w.r.t. $\odot O$. Since A and C are invariants under this inversion, and A, P, C are collinear, their images A, E, C together with O are concyclic by proposition 1.3.

Next, we consider another inversion with centre P and ratio $PA \cdot PC$ (in directed lengths). Since

$$PB \cdot PD = PA \cdot PC = PO \cdot PE,$$

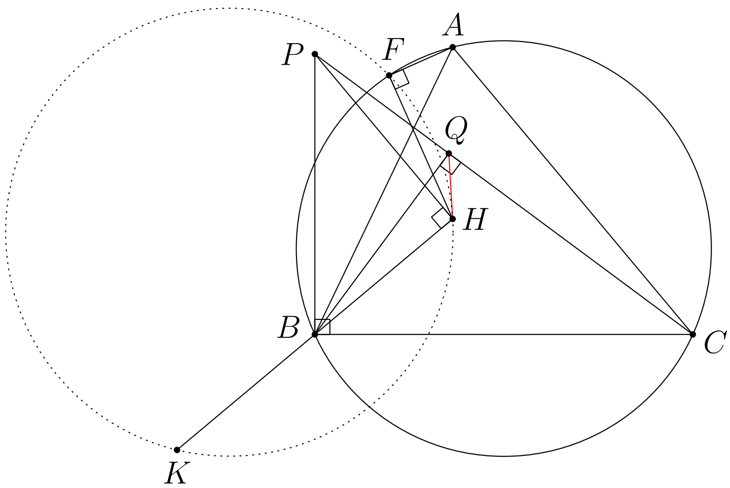
the following pairs of points are swapped: $(B, D), (A, C), (O, E)$. Therefore, (ABP) and (CDP) are mapped to the lines CD and AB respectively. This shows the image of Q is the intersection of these lines, which is T. Thus, T lies on PQ and

$$PT \cdot PQ = PA \cdot PC = PO \cdot PE.$$

Finally, this implies O, Q, E, T are concyclic, and hence $\angle OQP = \angle TEP = 90^\circ$. $\square$

Remarks. The above solution demonstrates the use of an inversion with negative ratio (the one with centre P). It does not differ too much from the usual inversion. Also, we use more than one inversion in the solution. This can be useful if the inversions do not greatly change the figures. This usually happens when there are many groups of concyclic points.

Problem 3.3. (Serbia TST 2018 Problem 5) Let H be the orthocentre of $\triangle ABC$, where $AB \neq AC$, and let F be a point on the circumcircle of $\triangle ABC$ such that $\angle AFH = 90°$. K is the symmetric point of H with respect to B. Let P be a point such that $\angle PHB = \angle PBC = 90°$, and Q is the foot of B to CP. Prove that HQ is tangent to the circumcircle of $\triangle FHK$.

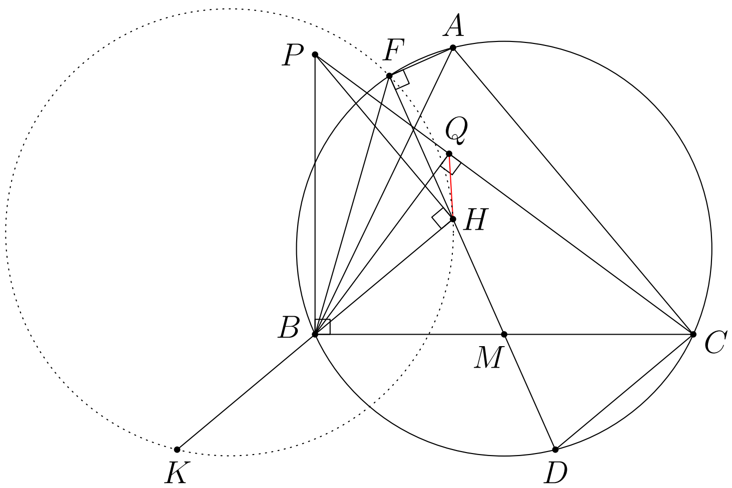


Analysis. When we first work on a geometry problem, we may not think of inversion directly. Let us first do some analysis in the original figure. Given the condition $\angle AFH = 90°$, it is natural to extend the line FH to a chord of (ABC). Indeed, the second intersection point of FH and (ABC) would be the point opposite to A on (ABC). One may also recall that the line FH passes through the midpoint M of BC if one is experienced in triangle geometry.

This time we are going to use an inversion which is commonly used in triangle geometry. Let A', B', C' be the feet of altitudes of $\triangle ABC$ from A, B, C respectively. As $HA \cdot HA' = HB \cdot HB' = HC \cdot HC'$, we can apply an inversion at centre H that swaps the pairs of vertices and the feet of altitudes. Using the above fact, it is not hard to see that F and M are swapped as well. Another hint of using this inversion is that (FKH) is mapped to a straight line while QH remains a straight line. Therefore, we only need to prove a pair of parallel lines afterwards, which looks simpler than the original problem.

The hardest step in this proof is to find the images of P and Q. The crucial observation is that P, F, Q, H, B are concyclic. If we are lucky enough, we may have figured this out before we attempt an inversion proof.

Solution.

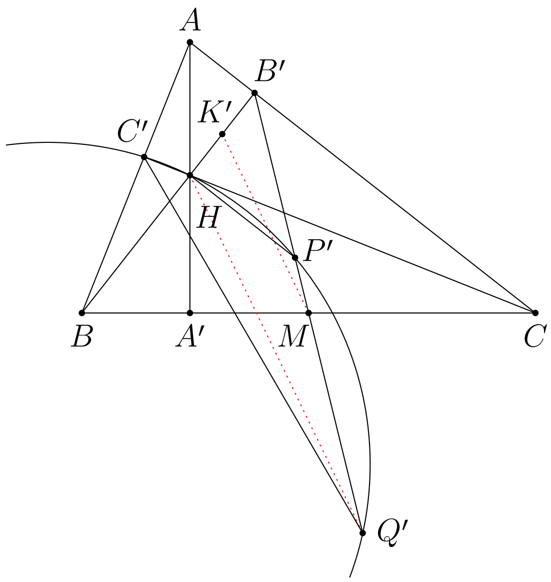


Let M be the midpoint of BC, and let D be the point opposite to A on (ABC). It is well-known that F, H, M, D are collinear. Note that

$$\angle BFH = \angle BFD = \angle BCD = 90^\circ - C = \angle CBH = 90^\circ - \angle HBP = \angle BPH.$$

Therefore, P, F, H, B are concyclic. In addition, $\angle PQB = 90^\circ = \angle PHB$. Thus, P, F, Q, H, B are concyclic.

Now, let A', B', C' be the feet of altitudes from A, B, C respectively. Consider the inversion at centre H and ratio $HA \cdot HA'$. Since $HA \cdot HA' = HB \cdot HB' = HC \cdot HC'$, the images of A, B, C are exactly A', B', C' respectively. Note that we also have $\angle AFM = 90^\circ = \angle AA'M$. This implies A, F, A', M are concyclic, and hence the image of F is M. Next, as B is the midpoint of HK and $HB \cdot HB' = HK \cdot HK'$, we know that K' is the midpoint of HB'.



The line PH is an invariant, while $(PFQHB)$ is mapped to the line $B'M$. Therefore, P' is the intersection point of PH and $B'M$. The line CP is mapped to $(HC'P')$. Thus, Q' is the second intersection point of $B'M$ and $(HC'P')$. Furthermore, the line QH becomes the line $Q'H$ (or remains unchanged), while (FHK) is mapped to the line MK'. This means we need to prove $Q'H//MK'$.

Recall that K' is the midpoint of HB'. Therefore, $Q'H//MK'$ iff M is the midpoint of $B'Q'$ by the midpoint theorem. The latter should be easier to prove because we do not need the point K'. As M is also the midpoint of BC, it suffices to show $B'BQ'C$ is a parallelogram (or indeed it should be a rectangle). Now,

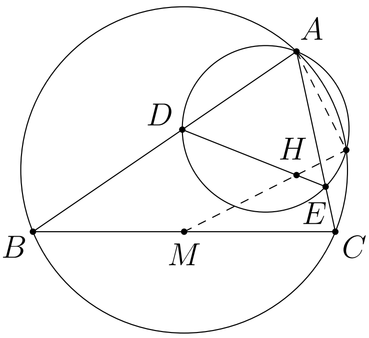
$$\angle B'Q'C' = \angle P'Q'C' = \angle P'HC = \angle B'CC'.$$

The last equality holds since $HP'//B'C$ (as both are perpendicular to BB'). This implies C, B', C', Q' are concyclic, and hence C, B', C', B, Q' are concyclic. The rest is simple:

$$\begin{aligned} \angle Q'BC = \angle Q'B'C = \angle B'CB \quad &\Rightarrow \quad BQ'//B'C, \\ \angle CQ'B' = \angle CBB' = \angle BB'Q' \quad &\Rightarrow \quad CQ'//B'B. \end{aligned}$$

This shows $B'BQ'C$ is a parallelogram as desired. $\square$

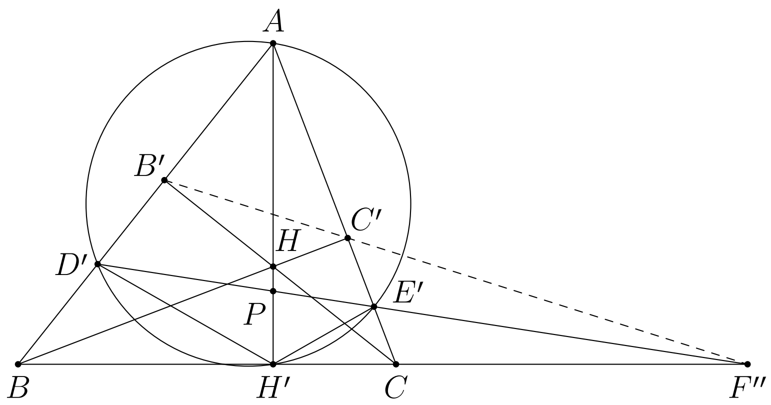
Problem 3.4. (IMO Shortlist 2005 G5) Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocentre of $\triangle ABC$, and let M be the midpoint of the side BC. Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of $\triangle ABC$ and $\triangle ADE$.



Analysis. Again, recall that the ray MH intersects the circumcircle at a point F such that $\angle AFH = 90°$. In particular it lies on the circle with diameter AH. If one knows this fact, then it is natural to add this new circle in the figure. Then there are three circles passing through A, and one may consider an inversion with centre A. We are going to introduce an inversion which is also useful in problems related to the orthocentre, other than the one used in the previous problem.

The following solution requires some knowledge in projective geometry. Alternatively, those steps can be replaced by some computational steps using trigonometric functions.

Solution.



Let F be the second intersection of (ABC) and (ADE). Let H', C', B' be the feet of altitudes from A, B, C respectively. Consider the inversion with centre A and ratio $AB \cdot AB'$. Since

$$AB \cdot AB' = AC \cdot AC' = AH \cdot AH',$$

the images of B, C, H are exactly B', C', H' respectively.

The line DE becomes a circle through A, and so A, D', H', E' are concyclic. As $AD = AE$, we have $AD' = AE'$ by the definition of inversion. Thus, AH' is the angle bisector of $\angle E'H'D'$.

Next, F' is the intersection of the lines $B'C'$ and $D'E'$, as these are the images of the two given circles. To show that $\angle AFH = 90^\circ$ (which is our goal), it is the same as showing F lies on $(AB'HC')$ as this circle has AH as a diameter. Equivalently, we have to show that B, C, F' are collinear.

Rather than proving this directly, we define F'' to be the intersection point of the lines BC and $D'E'$. Also, let P be the intersection point of AH' and $D'E'$. As $\angle PH'F'' = 90^\circ$ and $H'P$ bisects $\angle D'H'E'$, we have $(D', E'; P, F'') = -1$ (which is a well-known fact of harmonic division). Consider the following projection.

$$D'E'(D', E', P, F'') \xrightarrow{A} BC(B, C, H', F'')$$

This shows $(B, C; H', F'') = -1$. Since the cevians AH', BC', CB' are concurrent, the points B', C', F'' are collinear by another well-known fact of harmonic division. This proves $F'' = F'$, and so we are done. $\square$

Remarks. For more details of projection and harmonic division, one may refer to the note **Projective Geometry 4.0**.

Next, we are going to introduce an extremely useful inversion in triangle geometry. This can be used to solve many problems, especially when the incentre and excentres are involved.

Given a $\triangle ABC$, consider an inversion at centre A and radius $\sqrt{AB \cdot AC}$ (or ratio $AB \cdot AC$), together with a reflection in the angle bisector of $\angle BAC$. This is known as the $\sqrt{bc}$-inversion because b and c usually denote the lengths of AC and AB respectively. Under this inversion, the points B and C are swapped with each other (to be explained in the problem below). As before, the advantage of doing so is that one can compare the new figure with the original one.

We first demonstrate how to use the $\sqrt{bc}$-inversion to prove several well-known results for the mixtilinear incircle. Recall that the A-mixtilinear incircle means the circle lying inside (ABC) which is tangent to AB, AC and (ABC).

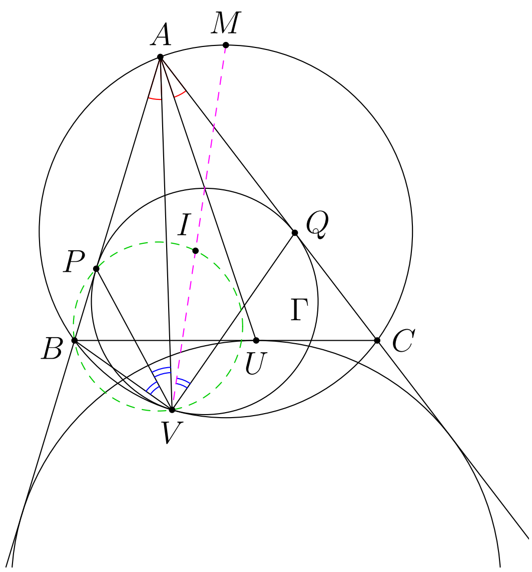
Problem 3.5. In $\triangle ABC$, let Γ be the A-mixtilinear incircle. Suppose Γ is tangent to AB, AC and (ABC) at P, Q and V respectively. Let I be the incentre of $\triangle ABC$, let U be the contact point of the A-excircle with BC, and let M be the midpoint of $\widehat{BAC}$ on (ABC). Prove the following results.

(a) $\angle BAV = \angle CAU$

(b) $\angle BVP = \angle AVP = \angle IVQ$

(c) B, P, I, V are concyclic

(d) V, I, M are collinear



Analysis. All these results can be proved by using the same inversion. We shall see how this can transform the mixtilinear incircle to something that is familiar to us.

Solution. Consider the inversion with centre A and ratio $AB \cdot AC$ together with the reflection in the angle bisector of $\angle BAC$. After the inversion, B is mapped to a point B' on AB such that $AB' = AC$. Therefore, after the reflection, B' is mapped to C. Similarly, the image of C is B.

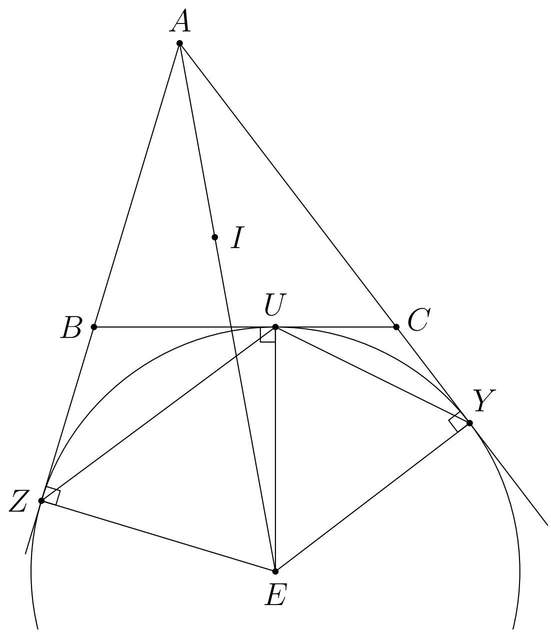
Note that the reflection does not change the shape of anything. Therefore, we can argue as usual. For example, the line BC is mapped to (ABC) because C and B are the images of B and C respectively. Conversely, (ABC) is mapped to the line BC. This is another useful property of this $\sqrt{bc}$-inversion.

Next, since Γ is tangent to AB, AC and (ABC), its image is tangent to AC, AB and BC. Note that the image of Γ lies inside $\angle BAC$ and outside $\triangle ABC$ (for example, as $AP < AB$, the image of P lies on the ray AC beyond C). Therefore, the image of Γ is the A-excircle. In particular, V (contact point of Γ and (ABC)) is mapped to U (contact point of the A-excircle and BC), while P and Q are mapped to the contact points Y and Z of the A-excircle and the lines AC and AB respectively.

In addition, the incentre I is mapped to the A-excentre E of $\triangle ABC$ since

$$AI \cdot AE = AB \cdot AC$$

and all of A, I, E lie on the internal angle bisector of $\angle BAC$. (This is another useful fact that is often used in this $\sqrt{bc}$-inversion.)



(a) The result $\angle BAV = \angle CAU$ follows immediately from the fact that V is mapped to U under the transformation, because it shows AV and AU are symmetric about the internal angle bisector of $\angle BAC$. $\square$

(b) We use this example to demonstrate how to find the relations between the angles before and after an inversion. Recall that angles with a label being the centre of inversion can be easily compared. Therefore, we write $\angle BVP = \angle AVB - \angle AVP$. As V and B are mapped to U and C respectively, we have $\angle AVB = \angle ACU$. Similarly, we have $\angle AVP = \angle AYU$. Thus,

$$\angle BVP = \angle ACU - \angle AYU = \angle CUY = \frac{C}{2}.$$

Also, we have

$$\angle AVP = \angle AYU = \frac{C}{2}$$

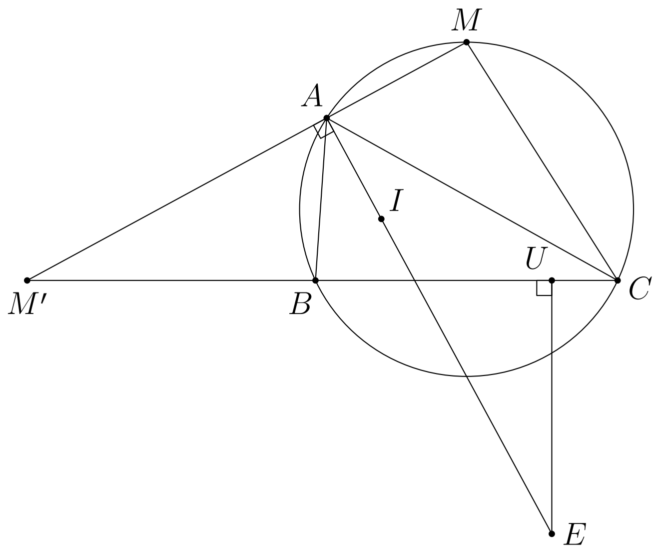
and

$$\begin{aligned}\angle IVQ &= \angle AVQ - \angle AVI = \angle AZU - \angle AEU = \angle AZU - (\angle ZEU - \angle AEZ)\\ &= \frac{B}{2} - \left(B - \left(90^\circ - \frac{A}{2}\right)\right) = 90^\circ - \frac{A}{2} - \frac{B}{2} = \frac{C}{2}.\end{aligned}$$

Therefore, all these angles are equal. □

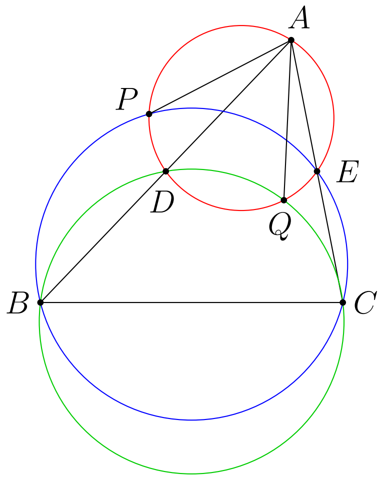
(c) The points B, P, I, V are mapped to C, Y, E, U respectively. These points are concyclic since $\angle EUC = \angle EYC = 90^\circ$. Therefore, B, P, I, V are concyclic. □

(d) To show that V, I, M are collinear, we have to show that the image of M lies on (AUE) since U and E are the images of V and I respectively. It is well-known that AM is the external angle bisector of $\angle BAC$. Therefore, $\angle MAE = 90^\circ$ and the image of M lies on the line AM. Let M' be the intersection of AM and BC. Then M' lies on (AUE) as $\angle M'AE = 90^\circ = \angle M'UE$. So it remains to show that M' is the image of M (which is also the reason of defining this point M').



Since M' lies on AM such that M and M' are separated by A, the only thing to check is $AM \times AM' = AB \times AC$ by definition. Indeed, using $\angle ABM' = \angle AMC$ and $\angle M'AB = 90^\circ - \frac{A}{2} = \angle CAM$, we have $\triangle AM'B \sim \triangle ACM$. Finally, this implies $\frac{AM'}{AB} = \frac{AC}{AM}$, and the proof is complete. □

Problem 3.6. (CGMO 2014 Problem 6) In acute $\triangle ABC$, $AB > AC$. D and E are the midpoints of AB, AC respectively. The circumcircle of $\triangle ADE$ intersects the circumcircle of $\triangle BCE$ at P (distinct from point E). The circumcircle of $\triangle ADE$ intersects the circumcircle of $\triangle BCD$ at Q (distinct from point D). Prove that $AP = AQ$.



Analysis. Although there is no mixtilinear incircle, incentre or excentre in the figure, the $\sqrt{bc}$-inversion is still useful in solving this problem. Instead of swapping B and C, we choose to swap B and E instead. This is just for convenience as the only effect is to reduce the size of the figure.

Solution. Consider the inversion with centre A and ratio $AB \cdot AE$ together with the reflection in the angle bisector at A. Then the image of B is E and vice versa. Similarly, C and D are interchanged after the transformation since $AB \cdot AE = AC \cdot AD$.

The circumcircle of $\triangle ADE$ becomes the line BC. The circumcircle of $\triangle BCE$ becomes (EDB), while the circumcircle of $\triangle BCD$ becomes (EDC). Thus, the image P' of P is the second intersection point of BC and (BDE), the image Q' of Q is the second intersection point of BC and (CDE).

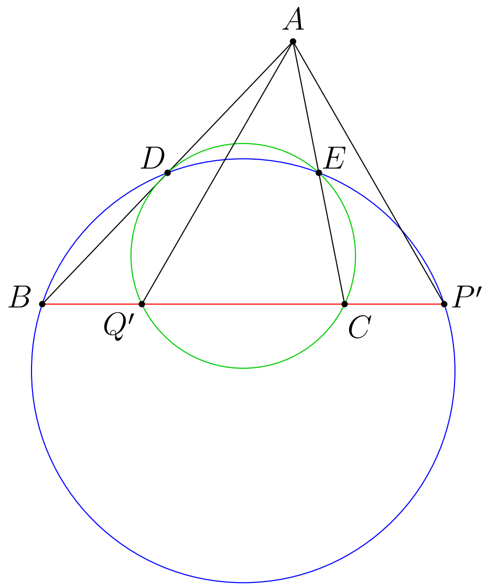
Note that $AP' = \frac{AB \cdot AE}{AP}$ and $AQ' = \frac{AB \cdot AE}{AQ}$. So it suffices to show $AP' = AQ'$. But this is simple. Indeed, as $DE // BC$, both $EDBP'$ and $EDQ'C$ are isosceles trapezoids. If we let D', A', E' be the projections of D, A, E onto BC respectively, then

$$Q'A' = Q'D' + D'A' = E'C + D'A' = A'E' + D'A' = D'E'$$

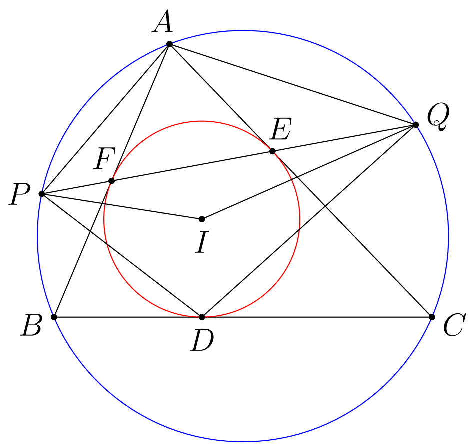
and

$$A'P' = A'E' + E'P' = A'E' + BD' = A'E' + D'A' = D'E' = Q'A'.$$

Therefore, $AP' = AQ'$, and hence $AP = AQ$ as desired. $\square$



Problem 3.7. (IMO Shortlist 2019 G6) Let I be the incentre of acute-angled triangle ABC. Let the incircle meet BC, CA, and AB at D, E, and F, respectively. Let line EF intersect the circumcircle of the triangle at P and Q, such that F lies between E and P. Prove that $\angle DPA + \angle AQD = \angle QIP$.



Analysis. Even if we are going to solve this problem by inversion, it is not clear which inversion we should use. It does not seem there is any inversion that can simplify the angle relation to be shown. The first idea is just to attempt anything that we may think of. This often happens for a hard problem. In this solution, we first try to do an inversion w.r.t. the incircle.

Solution. Firstly, consider an inversion w.r.t. the incircle of $\triangle ABC$ (with centre I). The points D, E, F are invariants as they lie on the incircle. The points A, B, C are mapped to the midpoints of EF, FD, DE respectively. Therefore, (ABC) is mapped to the nine-point circle of $\triangle DEF$. Next, the line EF is mapped to (EFI). This implies P' and Q' are the intersection points of $(A'B'C')$ and (EFI).

We now obtain an equivalent statement to $\angle DPA + \angle AQD = \angle QIP$ in the new figure. We have

$$\angle DPA = \angle IPD + \angle IPA = \angle IDP' + \angle IA'P'$$

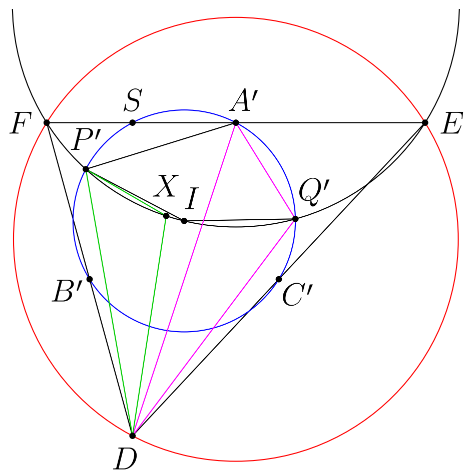
and similarly $\angle AQD = \angle IDQ' + \angle IA'Q'$. This shows

$$\angle DPA + \angle AQD = (\angle IDP' + \angle IDQ') + (\angle IA'P' + \angle IA'Q') = \angle P'DQ' + \angle P'A'Q'.$$

Also, it is clear that $\angle QIP = \angle P'IQ'$. Therefore, the goal is to prove

$$\angle P'DQ' + \angle P'A'Q' = \angle P'IQ'.$$

This does not seem to simplify the whole problem, but in fact, this goal can be expressed in terms of a much nicer geometric fact by introducing an important point.



Let X be the point such that $\triangle DXP'$ and $\triangle DQ'A'$ are directly similar. By spiral similarity, $\triangle DXQ'$ and $\triangle DP'A'$ are also directly similar. It follows that

$$\begin{aligned}\angle P'DQ' + \angle P'A'Q' &= \angle P'DQ' + (\angle P'A'D + \angle Q'A'D)\\ &= \angle P'DQ' + \angle XQ'D + \angle XP'D\\ &= \angle P'XQ'.\end{aligned}$$

Now the goal can be simplified to $\angle P'XQ' = \angle P'IQ'$, which means we have to show P', X, I, Q' are concyclic.

Here is the step to apply another inversion. We apply the inversion with centre D and ratio $DB' \cdot DE$ together with a reflection in the angle bisector of $\angle EDF$. This inversion swaps the pairs (E, B'), (F, C') as before. In addition, as I is the circumcentre of $\triangle DEF$, if we let S be the foot of altitude from D, then I and S are swapped with each other because $\triangle DB'S \sim \triangle DIE$ (hence $\angle B'DS = \angle EDI$ and $DS \cdot DI = DB' \cdot DE$). Therefore, (EFI) is mapped to $(B'C'S)$, which is exactly the nine-point circle of $\triangle DEF$. This is indeed the major reason why we would apply this inversion.

As we know that (EFI) and $(A'B'C')$ are swapped with each other, their intersection points P' and Q' also swap with each other. Thus, DP' and DQ' are isogonals w.r.t. $\angle EDF$, and $DP' \cdot DQ' = DB' \cdot DE$.

Recall that $\triangle DXP'$ and $\triangle DQ'A'$ are directly similar. Therefore, DX and DA' are isogonals w.r.t. $\angle EDF$, and $DX \cdot DA' = DP' \cdot DQ' = DB' \cdot DE$. This shows X and A' are swapped with each other. As A' lies on the nine-point circle, its image X must lie on (EFI) as desired. □

Remarks. The point X is indeed the D-dumpty point of $\triangle DEF$ as it lies on the D-symmedian of $\triangle DEF$ and (EFI).

The above solution shows another situation that the $\sqrt{bc}$-inversion might be useful. When there is a pair of directly similar triangles sharing the same vertex (like $\triangle DXP'$ and $\triangle DQ'A'$), there exists an inversion together with a reflection that can swap two pairs of vertices.

Here is the summary of the tricks used in this section.

- swapping pairs of points using concyclic points (problems 3.1, 3.2, 3.3, 3.4)
- applying inversions related to the orthocentre (problems 3.3, 3.4)
- applying the $\sqrt{bc}$-inversion (problems 3.5, 3.6, 3.7)
- swapping pairs of points using similar triangles (problem 3.7)

4 Changing the Lengths

In some geometry problems, there might be some conditions or results to be proved that are expressed in terms of lengths. It is possible that after applying a certain inversion, the lengths can be easier to compare or the expressions become much nicer. We are going to study some problems of this kind in this section.

Problem 4.1. (Ptolemy's inequality) Let $ABCD$ be a convex quadrilateral. Prove that

$$AB \cdot CD + BC \cdot DA \geqslant AC \cdot BD,$$

and equality holds iff $ABCD$ is cyclic.

Analysis. This is a well-known application of inversion. The theorem is trivialized by considering an inversion at any vertex.

Solution. Consider an inversion at centre A and ratio k. By proposition 1.4, we have

$$AB \cdot CD = AB \cdot \left(\frac{AC \cdot AD}{k} \cdot C'D'\right) = \frac{AB \cdot AC \cdot AD}{k} \cdot C'D'.$$

Similarly,

$$BC \cdot DA = \frac{AB \cdot AC \cdot AD}{k} \cdot B'C'$$

and

$$AC \cdot BD = \frac{AB \cdot AC \cdot AD}{k} \cdot B'D'.$$

The desired inequality is the same as $C'D' + B'C' \geqslant B'D'$. This obviously holds by the triangle inequality. Equality holds when B', C', D' are collinear (note that C' lies inside $\angle B'AD'$ as $ABCD$ is convex). Equivalently, A, B, C, D are concyclic. $\square$

Problem 4.2. (International Zhautykov Olympiad 2020 Problem 3) Given a convex hexagon $ABCDEF$ inscribed in a circle, prove that

$$AC \cdot BD \cdot CE \cdot DF \cdot EA \cdot FB \geqslant 27 \cdot AB \cdot BC \cdot CD \cdot DE \cdot EF \cdot FA.$$

Analysis. Applying an inversion w.r.t. the given circle will not change anything. On the other hand, applying an inversion with one of the given points as centre is a good idea because the circle will become a straight line. This gives us a new problem to work on.

Solution. Apply an inversion with centre A and ratio k. The points B, C, D, E, F are mapped to collinear points B', C', D', E', F' in this order. By proposition 1.4, we have

$$
\begin{aligned}
&AC \cdot BD \cdot CE \cdot DF \cdot EA \cdot FB \\
= {} & AC \cdot \left(\frac{AB \cdot AD}{k} \cdot B'D'\right) \cdot \left(\frac{AC \cdot AE}{k} \cdot C'E'\right) \cdot \left(\frac{AD \cdot AF}{k} \cdot D'F'\right) \\
&\cdot AE \cdot \left(\frac{AF \cdot AB}{k} \cdot F'B'\right) \\
= {} & \frac{1}{k^4} \cdot (AB \cdot AC \cdot AD \cdot AE \cdot AF)^2 \cdot (B'D' \cdot C'E' \cdot D'F' \cdot F'B')
\end{aligned}
$$

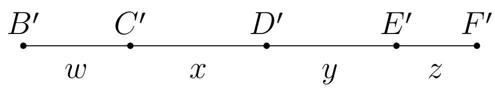
and

$$
\begin{aligned}
&AB \cdot BC \cdot CD \cdot DE \cdot EF \cdot FA \\
= {} & AB \cdot \left(\frac{AB \cdot AC}{k} \cdot B'C'\right) \cdot \left(\frac{AC \cdot AD}{k} \cdot C'D'\right) \cdot \left(\frac{AD \cdot AE}{k} \cdot D'E'\right) \\
&\cdot \left(\frac{AE \cdot AF}{k} \cdot E'F'\right) \cdot AF \\
= {} & \frac{1}{k^4} \cdot (AB \cdot AC \cdot AD \cdot AE \cdot AF)^2 \cdot (B'C' \cdot C'D' \cdot D'E' \cdot E'F').
\end{aligned}
$$

Comparing these, we have to prove

$$B'D' \cdot C'E' \cdot D'F' \cdot F'B' \geqslant 27(B'C' \cdot C'D' \cdot D'E' \cdot E'F').$$

Recall that B', C', D', E', F' lie on a straight line in this order.



Let $w = B'C'$, $x = C'D'$, $y = D'E'$ and $z = E'F'$. The desired inequality is the same as

$$(w+x)(x+y)(y+z)(w+x+y+z) \geqslant 27wxyz.$$

We first guess the equality case. Due to symmetry, we may focus on the case $w = z$ and $x = y$. Then we have to solve

$$4x(w+x)^3 = 27w^2x^2.$$

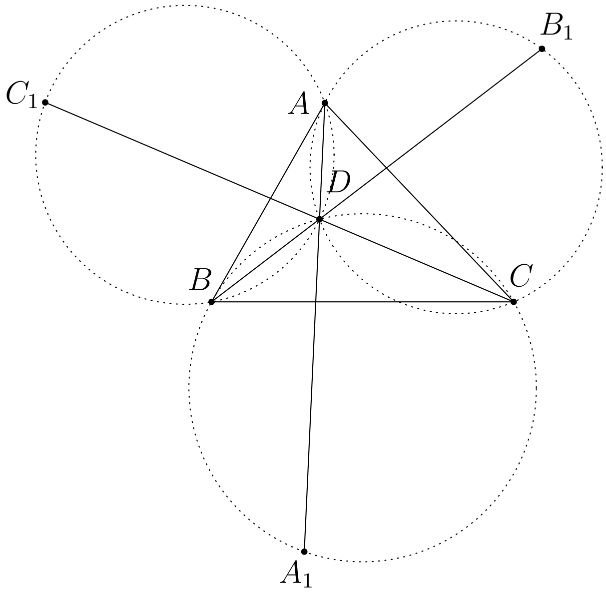
It is not hard to work out an equality case $w : x = 2 : 1$, i.e. $w : x : y : z = 2 : 1 : 1 : 2$. Because of this, we apply the AM-GM inequality in the general case as follows:

$$
\begin{aligned}
&(w+x)(x+y)(y+z)(w+x+y+z) \\
= {} & \left(\frac{w}{2} + \frac{w}{2} + x\right)(x+y)\left(y + \frac{z}{2} + \frac{z}{2}\right)\left(\frac{w}{2} + \frac{w}{2} + x + y + \frac{z}{2} + \frac{z}{2}\right) \\
\geqslant {} & 3\sqrt[3]{\frac{w^2x}{4}} \cdot 2\sqrt{xy} \cdot 3\sqrt[3]{\frac{yz^2}{4}} \cdot 6\sqrt[6]{\frac{w^2xyz^2}{16}} \\
= {} & 27wxyz.
\end{aligned}
$$

This proves the desired inequality. $\square$

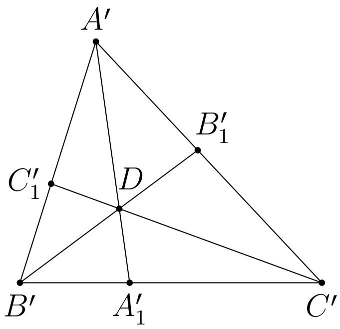
Problem 4.3. (Bay Area MO 2008 Grade 12 Problem 4) A point D lies inside triangle ABC. Let A_1, B_1, C_1 be the second intersection points of the lines AD, BD, and CD with the circumcircles of BDC, CDA, and ADB, respectively. Prove that

$$\frac{AD}{AA_1} + \frac{BD}{BB_1} + \frac{CD}{CC_1} = 1.$$



Analysis. Using inversion is a good option because we can apply an inversion at centre D to convert all circles to straight lines.

Solution.



Apply an inversion with centre D and ratio k. Since (BCD) is mapped to the line $B'C'$, the image of A_1 is the intersection point of $A'D$ and $B'C'$. Similarly, A_1', B_1', C_1' are points on $B'C', C'A', A'B'$ such that $A'A_1', B'B_1', C'C_1'$ are concurrent at D. By proposition 1.4, we have

$$\frac{AD}{AA_1} = AD \cdot \left(\frac{k}{DA \cdot DA_1} \cdot \frac{1}{A'A_1'} \right) = \frac{k}{DA_1} \cdot \frac{1}{A'A_1'} = \frac{DA_1'}{A'A_1'}.$$

Similarly, we need to prove

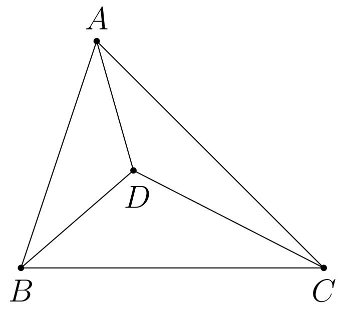
$$\frac{DA_1'}{A'A_1'} + \frac{DB_1'}{B'B_1'} + \frac{DC_1'}{C'C_1'} = 1.$$

This is a rather simple fact. Indeed, using areas, we obtain

$$\frac{DA_1'}{A'A_1'} + \frac{DB_1'}{B'B_1'} + \frac{DC_1'}{C'C_1'} = \frac{[DB'C']}{[A'B'C']} + \frac{[DC'A']}{[A'B'C']} + \frac{[DA'B']}{[A'B'C']} = \frac{[A'B'C']}{[A'B'C']} = 1.$$

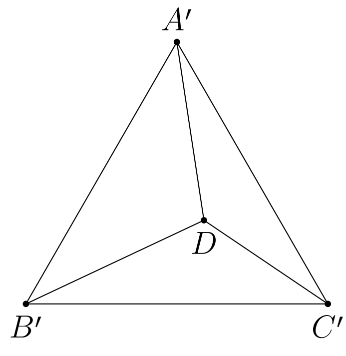
□

Problem 4.4. (Singapore MO Open 2016 Problem 1) Let D be a point in the interior of $\triangle ABC$ such that $AB = ab$, $AC = ac$, $BC = bc$, $AD = ad$, $BD = bd$, $CD = cd$. Show that $\angle ABD + \angle ACD = 60°$.



Analysis. The problem seems rather hard if we wish to solve it using trigonometry. What will happen if we apply an inversion? To add $\angle ABD$ and $\angle ACD$, we can apply an inversion at centre A or centre D.

Solution.



Apply an inversion at centre D and ratio k. By proposition 1.4, we have

$$A'B' = \frac{k}{DA \cdot DB} \cdot BA = \frac{k}{ad \cdot bd} \cdot ab = \frac{k}{d^2}.$$

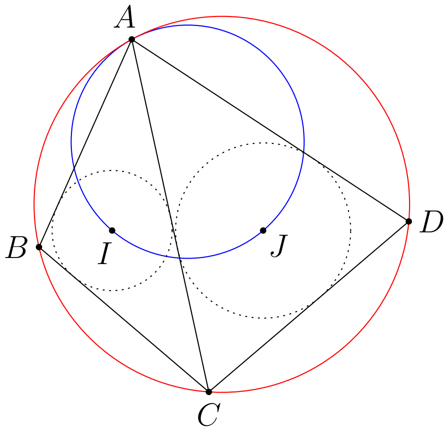
By symmetry, $B'C' = C'A' = \dfrac{k}{d^2}$. Therefore, $\triangle A'B'C'$ is equilateral, and hence

$$\angle ABD + \angle ACD = \angle DA'B' + \angle DA'C' = \angle B'A'C' = 60°.$$

□

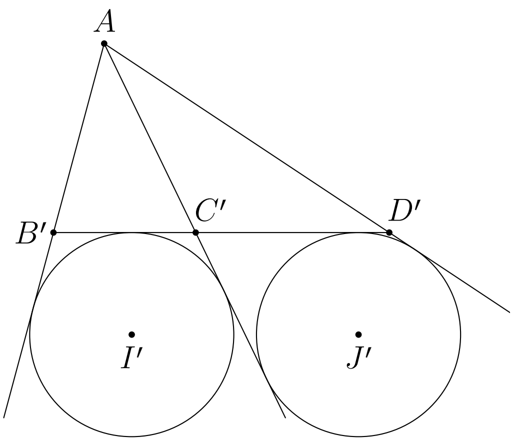
Problem 4.5. Given a cyclic quadrilateral $ABCD$, let I and J be the incentres of $\triangle ABC$ and $\triangle ADC$ respectively. Prove that the circumcircle of $\triangle AIJ$ is tangent to the circumcircle of $ABCD$ iff

$$\frac{CB}{CD} = \frac{AB + AC}{AD + AC}.$$



Analysis. If we apply an inversion at A, the tangential circles will become a pair of parallel lines. This makes it easier to compare the lengths.

Solution.



Consider an inversion with centre A and ratio k. Since A, B, C, D are concyclic, B', C', D' are collinear. Note that I is mapped to the A-excentre of $\triangle AB'C'$ as similar to the $\sqrt{bc}$-inversion (alternatively, this follows from $\angle B'AI' = \angle C'AI'$ and $\angle AI'B' = \angle ABI = \frac{1}{2}\angle ABC = \frac{1}{2}\angle AC'B'$). Analogously, J is mapped to the A-excentre of $\triangle AD'C'$.

Since $\triangle ABC \sim \triangle AC'B'$, we have

$$\frac{AB + AC}{CB} = \frac{AB}{CB} + \frac{AC}{CB} = \frac{AC'}{B'C'} + \frac{AB'}{B'C'} = \frac{AB' + AC'}{B'C'},$$

and similarly, $\dfrac{AD+AC}{CD} = \dfrac{AD'+AC'}{D'C'}$. Also, $\triangle AIJ$ and (ABD) are mapped to the lines $I'J'$ and $B'D'$ respectively. Therefore, (AIJ) is tangent to (ABD) iff $I'J'//B'D'$. This holds iff the A-exradius r_1 of $\triangle AB'C'$ is equal to the A-exradius r_2 of $\triangle AD'C'$. Now,

$$\begin{aligned}
& r_1 = r_2 \\
\Leftrightarrow \quad & \frac{2[AB'C']}{AB'+AC'-B'C'} = \frac{2[AD'C']}{AD'+AC'-D'C'} \\
\Leftrightarrow \quad & \frac{B'C'}{AB'+AC'-B'C'} = \frac{D'C'}{AD'+AC'-D'C'} \\
\Leftrightarrow \quad & \frac{B'C'}{AB'+AC'} = \frac{D'C'}{AD'+AC'} \\
\Leftrightarrow \quad & \frac{CB}{AB+AC} = \frac{CD}{AD+AC} \\
\Leftrightarrow \quad & \frac{CB}{CD} = \frac{AB+AC}{AD+AC}.
\end{aligned}$$

So we are done. $\square$

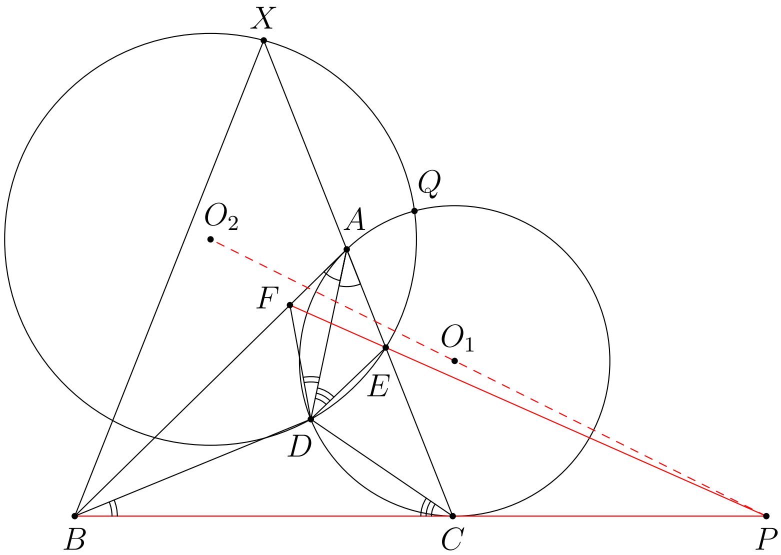
Problem 4.6. (IMO Shortlist 2021 G7/Problem 3) Let D be an interior point of the acute triangle ABC with $AB > AC$ so that $\angle DAB = \angle CAD$. The point E on the segment AC satisfies $\angle ADE = \angle BCD$, the point F on the segment AB satisfies $\angle FDA = \angle DBC$, and the point X on the line AC satisfies $CX = BX$. Let O_1 and O_2 be the circumcentres of the triangles ADC and EXD, respectively. Prove that the lines BC, EF, and O_1O_2 are concurrent.

Analysis. The following is a more computational proof. Among the three lines BC, EF and O_1O_2. It seems like BC and EF are simpler since E and F can be defined without the point X. Therefore, we may first focus on the points A, B, C, D, E, F and look for some more properties of the intersection point P of BC and EF. In particular, we may see if it is possible to locate the point P in terms of some length conditions. The advantage is that we do not need to use E and F to define P. Menelaus' theorem is a suitable tool for this purpose.

Next, to show that P lies on O_1O_2, it may be more natural to define the other intersection point Q of (ADC) and (EXD). Since O_1O_2 is the perpendicular bisector of DQ, it suffices to show P also lies on this perpendicular bisector, or equivalently $PD = PQ$. This slightly simplifies the problem because O_1 and O_2 are not needed anymore (while only one additional point is defined).

Inversion is used in some steps of the proof to simplify the figure.

Solution.



Let P be the intersection point of BC and EF. By Menelaus' theorem and the ratio lemma, we have

$$\frac{BP}{PC} = \frac{EA}{CE} \times \frac{FB}{AF} = \frac{AD \sin \angle ADE}{CD \sin \angle CDE} \times \frac{BD \sin \angle BDF}{AD \sin \angle ADF}.$$

Note that $\sin \angle CDE = \sin \angle BDF$ since

$$\begin{aligned} \angle CDE + \angle BDF &= 360^\circ - \angle BDC - \angle ADF - \angle ADE \\ &= 360^\circ - \angle BDC - \angle CBD - \angle BCD \\ &= 180^\circ. \end{aligned}$$

It follows that

$$\frac{BP}{PC} = \frac{BD}{CD} \times \frac{\sin \angle ADE}{\sin \angle ADF} = \frac{BD}{CD} \times \frac{\sin \angle BCD}{\sin \angle CBD} = \frac{BD^2}{CD^2}.$$

We have successfully located the point P solely in terms of $\triangle ABC$ and the point D.

A crucial observation is that $\dfrac{BP}{PC} = \dfrac{BD^2}{CD^2}$ actually means $\triangle PCD \sim \triangle PDB$ (or equivalently, PD is tangent to (BCD)). This can be proved via false position. Alternatively, we provide a proof using inversion.

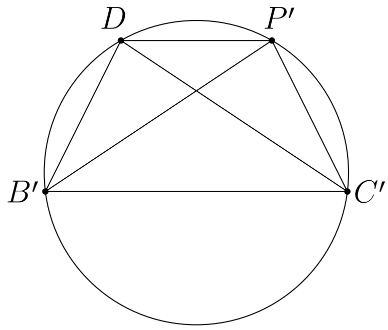
Apply an inversion at centre D and ratio k. By proposition 1.4, we have

$$\begin{aligned} BP &= \frac{DB \cdot DP}{k} \cdot B'P', \\ CP &= \frac{DC \cdot DP}{k} \cdot C'P', \end{aligned}$$

and so

$$\frac{BP}{PC} \cdot \frac{CD^2}{BD^2} = \frac{CD}{BD} \cdot \frac{B'P'}{C'P'} = \frac{B'D}{C'D} \cdot \frac{B'P'}{C'P'}$$

(since $\triangle DBC \sim \triangle DC'B'$). The condition becomes $\dfrac{B'D}{C'D} = \dfrac{C'P'}{B'P'}$. Note that $DB'C'P'$ is a cyclic quadrilateral since B, C, P are collinear in this order. Thus, we have $\angle B'DC' = \angle C'P'B'$. Together with $B'C' = C'B'$, we see that $\triangle B'DC' \cong \triangle C'P'B'$, and hence $DB'C'P'$ is an isosceles trapezoid. This proves $DP' // B'C'$, which means DP is tangent to (BCD).



Let Q be the second intersection point of (ADC) and (EXD). Then P lies on O_1O_2 iff $PD = PQ$. As $PD^2 = PB \times PC$, we need to prove $PQ^2 = PB \times PC$. Following the same steps as above, there are several equivalent statements that we can prove, such as the following:

- PQ is tangent to (BCQ);
- $\triangle PCQ \sim \triangle PQB$;
- $\dfrac{BP}{PC} = \dfrac{BQ^2}{CQ^2}$.

Among these, the last one looks easier to prove because we already know the position of P in terms of lengths. Indeed, comparing with $\dfrac{BP}{PC} = \dfrac{BD^2}{CD^2}$, it suffices to prove $\dfrac{BD}{CD} = \dfrac{BQ}{CQ}$. Now the point P is not needed.

The result $\dfrac{BD}{CD} = \dfrac{BQ}{CQ}$ looks like the angle bisector theorem. But this is not the case because neither D nor Q lies on BC, and neither B nor C lies on DQ. Can we change some of these points to collinear points? Yes, via inversion! If we choose a point that lies on the circumcircle of three of these points, then these three points are mapped to collinear points. Without defining extra points, an obvious choice of the centre would be A. Therefore, we are going to apply an inversion with centre A and any ratio. For simplicity, we still use B', etc. to mean the image of B. However, note that this is different from the B' we use in the above inversion.

After this inversion, D', C', Q' are collinear. But then what will the result be? It is useless if it becomes something much more complicated. The good news is that this actually does not change anything. Indeed, using similar triangles, we obtain

$$\frac{BD}{CD} = \frac{\frac{BD}{AD}}{\frac{CD}{AD}} = \frac{\frac{B'D'}{AB'}}{\frac{C'D'}{AC'}} = \frac{B'D'}{C'D'} \cdot \frac{AC'}{AB'}.$$

Similarly, $\frac{BQ}{CQ} = \frac{B'Q'}{C'Q'} \cdot \frac{AC'}{AB'}$. Therefore, it remains to prove

$$\frac{B'D'}{C'D'} = \frac{B'Q'}{C'Q'}.$$

In other words, we have to show that $B'C'$ bisects $\angle D'B'Q'$.

We need to find some more properties of the points D' and Q' to prove the result. Given A, B', C', the only condition on D' is that AD' is the angle bisector of $\angle B'AC'$. For the point Q', apart from lying on $D'C'$, another information should come from the fact that Q lies on (EXD). Thus, we first obtain some information about X'. Now,

$$\begin{aligned}
& \angle XBC = \angle XCB \\
\Rightarrow \quad & \angle ABX + \angle ABC = \angle ACB \\
\Rightarrow \quad & \angle AX'B' + \angle AC'B' = \angle AB'C' \\
\Rightarrow \quad & 180^\circ - \angle C'B'X' = \angle AB'C'.
\end{aligned}$$

Thus, X' is the point on the line AC' such that $B'C'$ is the external angle bisector of $\angle AB'X'$. Next, we have

$$\angle XQD = \angle XED = \angle CDE + \angle ECD = \angle CDE + \angle AQD.$$

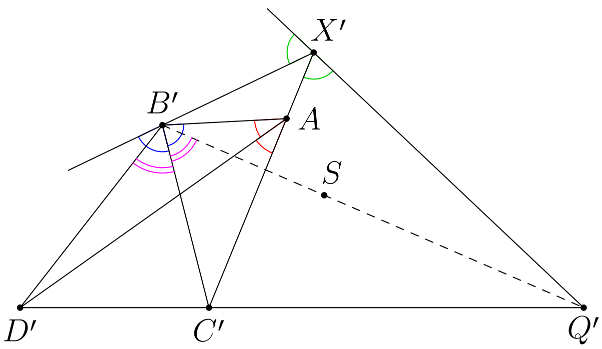
This implies $\angle XQA = \angle CDE$. We wish to obtain an expression not in terms of E, so that it is not necessary to find E'. This can be done as follows:

$$\angle XQA = \angle CDE = 180^\circ - \angle CAD - \angle ACD - \angle ADE = 180^\circ - \angle CAD - \angle ACB.$$

After inversion, this yields

$$\begin{aligned}
\angle AX'Q' &= 180^\circ - \angle C'AD' - \angle AB'C' \\
&= 180^\circ - \left(90^\circ - \frac{1}{2}\angle X'AB'\right) - \left(90^\circ - \frac{1}{2}\angle X'B'A\right) \\
&= \frac{\angle X'AB' + \angle X'B'A}{2} \\
&= 90^\circ - \frac{1}{2}\angle AX'B'.
\end{aligned}$$

Therefore, Q' is the intersection point of the external angle bisector of $\angle AX'B'$ and the line $C'D'$.



We have successfully changed the problem to an entirely different one. Now the only task is to prove that $B'C'$ bisects $\angle D'B'Q'$ using the equal angles given in the figure. Let S be a point such that $B'C'$ bisects $\angle D'B'S$. We have to prove $X'Q', B'S, D'C'$ are concurrent at Q'. One way is to use Ceva's theorem. For example, using Ceva's theorem in $\triangle B'C'X'$, we obtain

$$\begin{aligned}
&\frac{\sin\angle B'X'Q'}{\sin\angle C'X'Q'} \times \frac{\sin\angle X'C'D'}{\sin\angle B'C'D'} \times \frac{\sin\angle C'B'S}{\sin\angle X'B'S} \\
&= \frac{\sin(180^\circ - \angle C'X'Q')}{\sin\angle C'X'Q'} \times \frac{\sin\angle AC'D'}{\sin\angle B'C'D'} \times \frac{\sin\angle C'B'D'}{\sin(180^\circ - \angle AB'D')} \\
&= \frac{\sin\angle AC'D'}{\sin\angle B'C'D'} \times \frac{\sin\angle C'B'D'}{\sin\angle AB'D'} \\
&= \frac{\sin\angle C'AD'}{\sin\angle B'AD'} \quad \text{(by Ceva's theorem applied in } \triangle AB'C'\text{)} \\
&= 1.
\end{aligned}$$

This proves $X'Q', D'C', B'S$ are concurrent as desired. □

Remarks. Although the proof looks lengthy and scary, this is indeed a solution obtained by one of our team members during the contest! We only provide more details and ideas in the above proof.

In this section, there is no particular trick introduced. We mainly see how inversion can be used together with some length chasing. The major facts we need are proposition 1.4 and the similar triangles arisen from the inversion.

Links

Propositions

- 1.1: Relation between inversion and pole and polar transformation
- 1.2: Concyclic points and similar triangles under an inversion
- 1.3: Images of lines and circles under an inversion
- 1.4: Distance between image points under an inversion

Problems

- 2.1: Choosing centre of inversion
- 2.2: Inversion with respect to the incircle
- 2.3: IMO Shortlist 2003 G4
- 2.4: Tangent circles
- 2.5: IMO Shortlist 2008 G3
- 2.6: IMO 1996 Problem 2
- 2.7: Iranian Geometry Olympiad 2021 Advanced Problem 2
- 3.1: Swapping points
- 3.2: CMO 1992 Problem 4
- 3.3: Serbia TST 2018 Problem 5
- 3.4: IMO Shortlist 2005 G5
- 3.5: Properties of the mixtilinear incircle
- 3.6: CGMO 2014 Problem 6
- 3.7: IMO Shortlist 2019 G6
- 4.1: Ptolemy's inequality
- 4.2: International Zhautykov Olympiad 2020 Problem 3
- 4.3: Bay Area MO 2008 Grade 12 Problem 4
- 4.4: Singapore MO Open 2016 Problem 1
- 4.5: Tangential circles and lengths
- 4.6: IMO 2021 Problem 3

Terminologies and Notations

- inversion

References

- Mathematical Excalibur
  (Available from: https://www.math.hkust.edu.hk/excalibur/v9_n2.pdf)

「回首向來蕭瑟處，歸去，也無風雨也無晴。」

《定風波》